

Background Cross-Section Comparison with Table I

Independent cross-check of coherent photon bremsstrahlung and DIS backgrounds for *Electron-Ion Collider as a Discovery Tool for Invisible Dark Bosons*.

WORKING TEMPLATE: our independent photon and DIS rates are not yet validated. The yellow cells must be populated before this is sent as a numerical cross-check.

A. Coherent photon background

m_phi [GeV]	Paper [pb]	Our estimate +/- stat. [pb]	Difference [%]	Within 10%?
0.010	0.58	Pending	--	--
0.032	0.58	Pending	--	--
0.100	0.58	Pending	--	--
0.316	0.58	Pending	--	--
1.000	0.53	Pending	--	--
1.585	0.53	Pending	--	--
2.512	0.41	Pending	--	--
3.981	0.19	Pending	--	--
5.000	0.11	Pending	--	--
6.310	0.081	Pending	--	--
10.000	0.081	Pending	--	--

B. DIS background

m_phi [GeV]	Paper [pb]	Our estimate +/- stat. [pb]	Difference [%]	Within 10%?
0.010	0.0310	Pending	--	--
0.032	0.0310	Pending	--	--
0.100	0.0310	Pending	--	--
0.316	0.0310	Pending	--	--
1.000	0.0520	Pending	--	--
1.585	0.0520	Pending	--	--
2.512	0.0520	Pending	--	--
3.981	0.0330	Pending	--	--
5.000	0.0204	Pending	--	--
6.310	0.0155	Pending	--	--
10.000	0.0155	Pending	--	--

Comparison definition. Difference [%] = 100 x (our value - paper value) / paper value. A result passes the working agreement criterion when the absolute difference is below 10%.
Paper assumptions. Photon values use a missed-photon probability epsilon = 10^-6. The DIS column is (1 - epsilon_veto) A sigma_DIS_eff with epsilon_veto = 95%. All values are effective cross sections after the mass-dependent Table I cuts.

Source: H. Davoudiasl and H. Liu, arXiv:2505.08871v3 (29 June 2026), Table I. The published columns are transcribed exactly; no independent numerical results are asserted here.